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Hypothesis

WEIGHTS LIVE ON A MANIFOLD

- Training trajectories and loss landscapes suggest useful parameters occupy smooth low-dimensional regions inside $\mathbb{R}^P$.
- PCA and t-SNE snapshots of CNN layers show distinct, smooth trajectories rather than exploration of the full space.
- Therefore P target weights may be generated by a nonlinear differentiable map $g: \mathbb{R}^d \rightarrow \mathbb{R}^P$ with d much smaller than P .

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Theory

MAPPING + SOLVABILITY

- The Mapping Theorem states that a C2 low-dimensional parameterization can approximate optimal weights with arbitrarily small bounded loss error.
- A second theorem establishes local solvability and a gradient-based global extension for additive weight modulation.
- The theory formalizes the manifold hypothesis but depends on smoothness, Lipschitz and local-convergence assumptions.

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Architecture

LATENT TO FULL WEIGHTS

- A trainable latent z modulates fixed orthogonally initialized mapping weights: $w_{ij} \leftarrow w_{ij} + \alpha z_i$.
- The mapping output is partitioned and reshaped into every target weight and bias tensor; the target network is feed-forward only.
- Gradients flow through the generated parameters back to z and modulation, not into the target network itself.

MAPPING NETWORKS

Mapping Networks

Training Large Target Networks through Low-Dimensional Weight Manifolds | 2026

Do not optimize a target network's millions of weights directly; learn a compact latent vector whose modulated, mostly fixed mapping network generates the full target parameter set.

weight manifold

hypernetwork

parameter efficiency

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Mapping loss

TASK + GEOMETRY

- $L_{\text{map}} = L_{\text{task}} + \lambda_{\text{st}} L_{\text{stab}} + \lambda_{\text{sm}} L_{\text{smooth}} + \lambda_{\text{al}} L_{\text{align}}$
- Stability penalizes output changes under latent noise; smoothness controls the mapping Jacobian.
- Alignment matches latent and dominant mapping-weight directions; learned coefficients balance all terms.

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Evidence

200X-500X FEWER TRAINABLES

- Fashion-MNIST reaches 93.91% with 2,072 trainables versus CNN1's 92.89% with 537,994; layer-wise training reaches 94.83%.
- Celeb-DF reaches 88.88% with 2,048 trainables; Cityscapes uses 8,192 for 97.92% pixel accuracy and 0.4623 mIoU.
- A mapped LSTM reduces MSE from 0.0035 with 12,961 parameters to 0.00061 with 2,048; ResNet50 fine-tuning is also competitive.

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Scaling & limits

- Single-latent training is simplest; layer-wise latent vectors cut mapping-memory requirements by roughly 10x for larger networks.
- Low-rank decomposition, pruning and quantization are orthogonal add-ons; grouped modulation extends the idea to fine-tuning.
- Full mapping weights are fixed but still stored during training, so large targets remain memory-heavy and LLM/LVM scaling is future work.